Week 2: The likelihood function, estimation, and the likelihood principle

F79MA Statistical Models A

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Definition of likelihood and the Maximum Likelihood Estimator

We have seen how the MME is usually easy and convenient to find, but does not always give efficient or sensible estimators. *Is there a better way to derive statistical estimator of unknown model parameters?* A method that gives good estimators for a wide range of problems? The answer is yes: the **method of maximum likelihood**.

Suppose we observe data $\underline{x}$ in our experiment, and let $f_X(\underline{x};\theta)$ be the value of the probability function (pdf or pmf) at $\underline{x}$ given the parameter θ. Consider this as a function of θ to define the likelihood:

$$L(\theta;\underline{x}) = f_X(\underline{x};\theta)\,.$$

Informally, this is "the probability of observing the particular data $\underline{x}$, considered as a function of θ". More precisely, it will be something proportional to this probability since the pdf is not a probability but is proportional to a probability.

We will usually just write $L(\theta)$ for the likelihood, not $L(\theta;\underline{x})$.

If $\underline{x} = (x_1,\ldots,x_n)$, where the x_i are independent, then $f_X(\underline{x};\theta)$ factorises as a product and we have

$$L(\theta;\underline{x}) = \prod_{i=1}^{n} f_i(x_i;\theta)\,.$$

Note that the x_i don't have to have the same distribution here (i.e., each f_i might be different). If the sample is independent and identically distributed, then

$$L(\theta;\underline{x}) = \prod_{i=1}^{n} f_{X_1}(x_i;\theta)\,,$$

where, without loss of generality, f_{X_1} to denotes the probability function of the first element of the sample.

Definition: The *maximum likelihood estimator* (MLE) for θ is the value of θ that maximises $L(\theta;\underline{X})$. We will usually write $\widehat{\theta}$ for the MLE of θ.

We may also use the notation $\widehat{\theta}(\underline{X})$ to denote explicitly that the estimator is a function of $\underline{X}$. When we observe an outcome $\underline{x}$ of $\underline{X}$, that leads to the maximum likelihood

estimate $\widehat{\theta}(\underline{x})$, which is an outcome of the estimator $\widehat{\theta}(\underline{X})$. Indeed, the estimator is a statistic of the random variable $\underline{X}$, so $\widehat{\theta}(\underline{X})$ is a random variable as well. When we study the estimation bias or MSE, we analyse how the 1st and 2nd moments of that random variable compare to θ.

Regarding the derivation of the MLE, it is usually easier in practice to maximise $l(\theta) = \log L(\theta)$ rather than $L(\theta)$ (see examples below). Since log is an increasing function, these are equivalent. We call $l(\theta)$ the log-likelihood.

Example 2.1

Let $\underline{x} = (x_1, x_2, \ldots, x_n)$ be a random sample from Poisson distribution with parameter (mean) μ. In particular, suppose $n = 2$ and $\underline{x} = (5, 7)$.

We will explore the likelihood, the log-likelihood, and the MLE.

Here

$$L(\mu) = \frac{\mu^5 e^{-\mu}}{5!} \cdot \frac{\mu^7 e^{-\mu}}{7!} = \frac{1}{5! \cdot 7!} \mu^{12} e^{-2\mu} \,,$$

and

$$l(\mu) = \ln L(\mu) = -\ln(5!7!) + 12 \ln \mu - 2\mu \,.$$

We can obtain the MLE analytically, from $l(\mu)$:

$$\frac{d}{d\mu} l(\mu) = 12/\mu - 2 \,.$$

We equate this to zero for the maximum and get $\widehat{\mu} = 6$.

We have to verify that this is maximum and not minimum or saddle point. Indeed,

$$\frac{d^2 l(\mu)}{d\mu^2} = -\frac{12}{\mu^2} < 0 \,.$$

For the general case of a sample $\underline{x}$ of size n:

$$L(\mu) = \prod_{i=1}^{n} \frac{\mu^{x_i} e^{-\mu}}{x_i!} = \frac{\mu^{\sum_{i=1}^n x_i} e^{-n\mu}}{\prod x_i!}$$

and

$$l(\mu) = \left(\sum_{i=1}^{n} x_i \right) \ln \mu - n\mu - \text{constant} \,.$$

Then

$$\frac{dl}{d\mu} = \frac{\sum x_i}{\mu} - n \,.$$

Equate this to zero and get $\widehat{\mu} = \overline{x}$.

Check: $\frac{d^2 l(\mu)}{d\mu^2} = -n \frac{\overline{x}}{\mu^2} < 0$, and this is indeed a maximum.

Invariance under transformation A statistical model can be parameterised in many ways, for example:

$$\begin{aligned} \text{Normal model:} \quad & \mathrm{N}(0,\sigma^2)\,,\ \mathrm{N}(0,\phi)\,,\ \mathrm{N}(0,\psi^{-1})\,. \\ \text{Exponential model:} \quad & \mathrm{Exp}(\lambda)\,,\ \mathrm{Exp}(1/\mu)\,. \end{aligned}$$

How do MLEs for σ, ϕ and ψ relate to each other? How do MLEs for λ and μ relate to each other?

Sometimes we wish to estimate a function of θ. For example, in a Pois(θ) model we may wish to estimate $\mathbf{P}(X=0) = e^{-\theta}$. How can we do this?

Invariance property of MLEs: If the function g is bijective (one-to-one), then the MLE of the parametrisation $\phi = g(\theta)$ is $\hat{\phi} = g(\widehat{\theta})$. Or, in other words, MLEs are invariant under bijective transformations!

The likelihood principle

The likelihood principle is one of the key ideas in statistical inference. It essentially says that the likelihood tells you all there is to know from the data x about the parameter θ.

Likelihood Principle: If x and y are two possible experimental outcomes such that $L(\theta;x)$ is proportional to $L(\theta;y)$:

$$L(\theta;x) = C(x,y)L(\theta;y)\,,$$

then the conclusions about θ drawn from x and y should be identical.

Example A

We may model the number of new customers arriving to a shop during a particular day as a Pois(λ) distribution (independent of other days). Suppose we count the number of new customers seen on three consecutive days. Consider two possible outcomes of this experiment: $\underline{x} = (8,4,7)$ and $\underline{y} = (5,5,9)$. The likelihoods from these two outcomes are proportional (check this!) and therefore these two outcomes give the same conclusion about λ. In particular, they lead to the same maximum likelihood estimate.

Sufficiency

Consider Example A above. You can convince yourself (using the likelihood principle) that the same conclusions about the parameter λ would be drawn from any random sample of size 3 that sums to 19. For this example, the sum of the observed number of new customers is a *sufficient statistic* for λ (informally, it "gives you the same information about λ as the entire sample $\underline{x}$").

Definition: For an experiment with outcome $\underline{X}$, a statistic $T = T(\underline{X})$ is *sufficient* for θ if the conditional distribution of $\underline{X}$ given the value of T does not depend on θ:

$$f_{\underline{X}|T}(\underline{x}|t(\underline{x}) = t) = h(\underline{x}) \qquad \text{for all } \theta\,.$$

This is the definition of a sufficient statistic. But in practice the easiest way to find a sufficient statistic is usually to use the following result:

The Factorisation Theorem: $T = T(\underline{X})$ is sufficient for θ if and only if

$$f_{\underline{X}}(\underline{x};\theta) = h_1(\underline{x})h_2(t(\underline{x}),\theta)\,,$$

for some functions h_1 and h_2. Notice that θ only interacts with the data $\underline{x}$ via the statistic $t(\underline{x})$, so two experiments that produce the same value for this statistic will carry the same information about θ, even if the experimental outcomes are very different w.r.t. other statistics. In other words, θ only influences the distribution of $\underline{X}$ via the statistic $T(\underline{X})$, so this is the only aspect of the distribution that is relevant for inferences about θ.

Example 2.2

Let $\underline{X}$ be a random sample from a Pois(μ) distribution. We will

(i) Use the definition of sufficiency to show that $\sum_{i=1}^n X_i$ is a sufficient statistic for μ.

(ii) Use the factorisation theorem to find a sufficient statistic for μ.

You can see that (i) is much harder than (ii): this is why the factorisation theorem is so useful!

(i)

$$L(\mu) = f_{\underline{X}}(\underline{x};\mu) = \frac{\mu^{\sum x_i} e^{-n\mu}}{\prod x_i!}.$$

Let $T = T(\underline{X}) = \sum_{i=1}^n X_i$. We know that $T \sim \text{Pois}(n\mu)$, and so the probability mass function of $T(\underline{X})$ is

$$f_T(t;\mu) = \frac{(n\mu)^t e^{-n\mu}}{t!}.$$

To use the definition for sufficiency, we need the conditional probability mass function of $\underline{X}$ given $T = t$:

$$f_{\underline{X}|T=t}(\underline{x}|t) = \frac{f_{\underline{X},T}(\underline{x},t)}{f_T(t)}.$$

But

$$f_{\underline{X},T}(\underline{x},t) = \mathbf{P}(\underline{X} = \underline{x} \text{ and } t(\underline{x}) = t) = \mathbf{P}(\underline{X} = \underline{x}) \cdot \mathbf{I}(t(\underline{x}) = t).$$

Then

$$f_{\underline{X}|T=t}(\underline{x}|t) = \frac{f_{\underline{X},T}(\underline{x},t)}{f_T(t)} = \frac{t!}{n^t \prod x_i!}\mathbf{I}(t(\underline{x}) = t),$$

since $\sum_{i=1}^n x_i = t$.

This expression does not involve μ, and so by the definition of a sufficient statistic, $\sum_{i=1}^n X_i$ is sufficient for μ.

Note that $\overline{X} = \frac{1}{n}T(\underline{X})$ is therefore also sufficient for μ.

(ii) We have

$$\begin{aligned} f_{\underline{X}}(\underline{x};\mu) &= \frac{\mu^{\sum_{i=1}^n x_i} e^{-n\mu}}{\prod x_i!} \\ &= \left(\frac{1}{\prod x_i!}\right) \cdot \left(\mu^{\sum x_i} e^{-n\mu}\right) \\ &= h_1(\underline{x}) \cdot h_2\left(\sum_{i=1}^n x_i, \mu\right). \end{aligned}$$

So, $\sum_{i=1}^n X_i$ (or, more generally, any one-to-one function of $\sum_{i=1}^n X_i$) is a sufficient statistic for μ.

Example 2.3

Consider a random sample from the exponential distribution Exp(λ) with pdf $f(x;\lambda) = \lambda e^{-\lambda x}$ for $x > 0$.

(i) Find a sufficient statistic for λ.

(ii) Find the MLE $\widehat{\lambda}$.

Solution:

(i)

$$\begin{aligned} f_{\underline{X}}(\underline{x};\lambda) &= \prod_{i=1}^n \lambda e^{-\lambda x_i} \\ &= \lambda^n e^{-\lambda \sum_{i=1}^n x_i} \\ &= 1 \cdot \lambda^n e^{-\lambda \sum_{i=1}^n x_i} \\ &= h_1(\underline{x}) h_2\left(\sum_{i=1}^n x_i, \lambda\right). \end{aligned}$$

So, $\sum_{i=1}^n X_i$ (or any one-to-one function of $\sum_{i=1}^n X_i$ – say $\overline{X}$) is a sufficient statistic for λ.

(ii)

$$L(\lambda) = f_{\underline{X}}(\underline{x};\lambda) = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

and

$$l(\lambda) = \log L(\lambda) = n \log \lambda - \lambda \sum_{i=1}^n x_i.$$

Then

$$\frac{dl}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i.$$

By equating this to zero, we get the estimate $\widehat{\lambda} = 1/\overline{X}$. Since $\frac{d^2 l}{d\lambda^2} = -n\lambda^{-2} < 0$, this is a maximum.

Example 2.4

Let $\underline{X}$ be a random sample from the uniform $\mathrm{U}(0, \theta)$ distribution.

(i) Find a sufficient statistic for θ.

(ii) Find the MLE $\widehat{\theta}$.

Solution:

(i) We know that

$$f_{\underline{X}}(\underline{x}; \theta) = \prod_{i=1}^{n} f(x_i; \theta) = \left(\frac{1}{\theta}\right)^n \quad \text{if } 0 < x_i < \theta \text{ for all } i$$

and $f_{\underline{X}}(\underline{x}; \theta) = 0$ otherwise. This can be rewritten as

$$f_{\underline{X}}(\underline{x}; \theta) = \left(\frac{1}{\theta}\right)^n \cdot \mathbf{I}(\min x_i > 0) \cdot \mathbf{I}(\max x_i < \theta).$$

where $\mathbf{I}$ denotes the indicator function, i.e. $\mathbf{I}(\min x_i > 0)$ is 1 if all $x_i > 0$ and is 0 otherwise.

Hence,

$$h_1(\underline{x}) = \mathbf{I}(\min x_i > 0) \quad \text{and} \quad h_2(\underline{x}, \theta) = \left(\frac{1}{\theta}\right)^n \cdot \mathbf{I}(\max x_i < \theta).$$

Thus, $T = X_{(n)} = \max X_i$ is a sufficient statistic for θ.

(ii) Furthermore,

$$L(\theta) = f_{\underline{X}}(\underline{x}; \theta) = \left(\frac{1}{\theta}\right)^n \cdot \mathbf{I}(\min x_i > 0) \cdot \mathbf{I}(\max x_i < \theta).$$

Clearly, if at least one of x_i's is non-positive, then something is wrong! But if all are positive, then there is no need for the differentiation, and a direct inspection of the likelihood reveals that the MLE is $\widehat{\theta} = X_{(n)}$.

Higher-dimensional settings

We say that the statistics $T_1, \ldots, T_r$ are *jointly sufficient* for $\theta_1, \ldots, \theta_k$ if the conditional distribution of $\underline{X}$ given $T_1, \ldots, T_r$ does not depend on $\theta_1, \ldots, \theta_k$.

The factorisation theorem also holds in this case.

Exercise: Let $\underline{X}$ be a random sample from a normal $\mathrm{N}(\mu, \sigma^2)$ distribution, so that (μ, σ^2). Can you show that $(\overline{X}, S^2)$ are jointly sufficient for (μ, σ^2)?

Example: Let $\underline{X}$ be a random sample from the Cauchy distribution with pdf

$$f(x;\theta) = \frac{1}{\pi\left[1 + (x-\theta)^2\right]}\,.$$

This gives the likelihood:

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\pi\left[1 + (X_i-\theta)^2\right]}\,.$$

We can't use this to find a simpler set of sufficient statistics than the whole sample $\underline{X}$.

Final remark on sufficient statistics

We have seen that a sufficient statistic $T(\underline{X})$ contains all the information about θ that is contained in the data $\underline{X}$.

Why should we record and analyse all the information in $\underline{X}$ if a more parsimonious or compact representation based on a sufficient statistic exists?

Suppose we model $\underline{X}$ as a random sample of size 5 from a $\mathrm{Pois}(\lambda)$ distribution. A sufficient statistic is $\sum_{i=1}^{5} X_i$, and this is all we may be tempted to record and analyse.

But, suppose we obtained the data $\underline{x} = (1, 21, 0, 26, 2)$. These suggest that the Poisson model is wrong! In fact, when using the likelihood principle, the MLE, the factorisation theorem, we are assuming that the model that we have postulated for the data is the correct one. In this case, $\underline{x}$ **carries information that suggests our model is incorrect**, as $\underline{x} = (1, 21, 0, 26, 2)$ is an unlikely Poisson sample.